Quest #79

Registration

CONFIDENTIAL COMPANY

Hiring Quest – Core Software Engineer (Fullstack + AI) @ FinTech Company

Challenge-based hiring quest with structured evaluation and real project outcomes.

Status: Registration **Registration:** April 30, 2026 **Submission:** May 3, 2026

META INFO

CATEGORY	STATUS
Registration	Registration
REGISTRATION DEADLINE	**SUBMISSION DEADLINE**
April 30, 2026	May 3, 2026

PRIZE

Top performers get hired with a paid contract and the opportunity to work on real-world projects.

QUEST BRIEF

We are a FinTech company serving one of Egypt's major e-commerce platforms. As we enter our next phase of growth, we are building a core in-house engineering team to drive technology forward - and this role is at the heart of it.

We're hiring a **Core Software Engineer (Founding Engineer) (1–3 YOE)** to help us build this system from the ground up.

Start Date: Immediate
Location: **The City Center New Cairo, Egypt (On-site)**
Salary: **Starting From 25 K EGP**

How the Hiring Quest Works

- Register
- Submit your solution before the deadline
- Submissions are reviewed and all candidates receive feedback
- Top candidates join a technical review session to walk through their submitted task
- Qualified candidates are recommended to the hiring company for next steps

The technical session focuses on your thinking and understanding of your solution, so use AI as a support tool only — not a replacement for your own work.

Who We're Looking For

- 1–3 years of full-stack experience (Node.js or Python + React / Next.js)
- Strong backend + system design fundamentals
- Experience building APIs and working with databases
- Comfortable integrating AI/LLMs into products
- Has shipped real projects end-to-end

Bonus:

1. Vector databases (Pinecone, Chroma, FAISS)
2. Background jobs / queues
3. Search systems / retrieval pipelines
4. DevOps / Docker

Mindset:

1. Builder mindset (you create, not just implement)
2. High ownership & curiosity
3. Strong problem-solving and communication

Your Mission: "AI Knowledge Operations System"

Business Context

Modern teams are drowning in information:

1. Docs in Notion / Google Drive
2. Conversations in Slack
3. Data in dashboards
4. Decisions lost in meetings

Your mission is to build a system that:

Ingests knowledge from multiple sources
Structures it into searchable intelligence
Allows users to ask questions and get accurate, contextual answers
Surfaces insights proactively (not just reactively)

The Challenge

🌊 Step 1 – Multi-Source Knowledge Ingestion (Backend)

Build a system that:

- Accepts multiple data sources:
  - .pdf, .txt, .md
  - (Bonus: simulated Slack / Notion data as JSON)
- Processes:
  - Text extraction
  - Chunking
  - Embedding generation
- Stores:
  - Documents + metadata
  - Embeddings in a vector DB

Endpoints:

1. POST /api/ingest/files
2. POST /api/ingest/source (simulate external data)
3. GET /api/docs
4. GET /api/docs/:id

Bonus:

1. Background ingestion jobs
2. Deduplication
3. Versioning documents

🌊 Step 2 – Retrieval + Reasoning Engine (Core AI)

Build a **retrieval-augmented AI system**:

Endpoint:

POST /api/ai/query

Input:

```
{
 "question": "What decisions were made about product pricing last week?"
}
```

System should:

- Retrieve relevant chunks (semantic search)
- Rank / filter results
- Use LLM to:
  - Answer the question
  - Cite sources
  - Handle ambiguity

Output:

```
{
```

```
  "answer": "...",
  "sources": [...],
  "confidence": 0.91,
  "reasoning": "optional but bonus"
}
```

Bonus:

- Multi-step retrieval (query rewriting, re-ranking)
- Hybrid search (keyword + vector)
- Prompt engineering layers

🌊 Step 3 – AI Copilot Interface (Frontend)

Build a **knowledge assistant UI**, not just a chatbot.

Features:

- Upload & manage documents
- View processed knowledge base
- Ask questions in chat interface
- See:
  - Answers
  - Sources
  - Context snippets

UX Expectations:

- Clean, intuitive UI
- Fast responses
- Thoughtful interaction design

Tech:

- Next.js (App Router)
- TailwindCSS
- React Query / SWR

Bonus:

- Streaming responses
- Source preview panel
- Search + filters

🌊 Step 4 – Proactive Intelligence Layer (Advanced)

Move beyond Q&A → **AI that suggests insights**

Build a system that:

- Periodically scans knowledge base
- Generates insights like:
  - "Frequent issues mentioned in support logs"
  - "Repeated decisions across teams"
  - "Conflicting information detected"

Endpoint:

- GET /api/ai/insights

Bonus:

- Scheduled jobs
- Insight categorization
- Notifications (mocked)

🌊 Step 5 – System Design & Infrastructure

Design like a real product:

- Dockerized setup
- Modular architecture
- Logging & error handling

Bonus:

- Separate services (API / worker / AI service)
- Caching layer
- Rate limiting

Suggested Tech Stack

Backend: FastAPI / Node.js

Frontend: Next.js

Database: PostgreSQL

Vector DB: Chroma / FAISS / Pinecone

AI: OpenAI API

Infra: Docker Compose

Example Flow

User uploads documents → system processes + embeds

User asks question → retrieval + LLM reasoning

System returns answer + sources

Background job generates insights → shown in dashboard

Bonus Points

Advanced retrieval pipeline

Insight generation (proactive AI)

Clean system architecture

Strong UX thinking

Performance optimization

Observability

What You Should Submit

GitHub Repository (Production-Level)

- /backend (clean architecture)
- /frontend
- /infra
- /docs

Must include:

- README.md (setup + explanation)
- docker-compose.yml
- API documentation
- Database schema

If you used AI tools, please share your prompts and relevant conversation excerpts; submissions will be evaluated based on your understanding, not just the final output.

Strong Bonus:

- ARCHITECTURE.md:
  - System diagram
  - Data flow
  - Scaling approach
  - Trade-offs

15–20 Minute Deep-Dive Video

Part 1 (5 min) — You

- Your background
- Projects you built
- Technical challenges

Part 2 (10–12 min) — System Deep Dive

- Live demo (ingest → ask → insights)

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- Architecture explanation
- Retrieval + AI design decisions

Part 3 (3–5 min) — Founder Thinking

- How you scale this system
- What you'd improve next
- Biggest risks & trade-offs

Evaluation Criteria

System Design & Thinking 30%
AI / Retrieval Quality 25%
Backend Architecture 20%
Frontend UX & Product Thinking 15%
Code Quality 5%
Reliability 5%

After Submission

Top candidates will go through a **Founding Engineer Review Round**:

- Deep system design discussion
- AI architecture decisions
- Product thinking evaluation

Review process typically takes up to 10 working days after submission or review sessions.

QUEST SNAPSHOT

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Register Now

Need help? Contact us on WhatsApp at **01558405326** or join our **community**.

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